The Democratic Channel: An Information-Theoretic Measurement of Preference-Policy Transmission in the United States

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Abstract

We present, to our knowledge, the first empirical measurement of mutual information between citizen preferences and policy outcomes in the United States, using Shannon's information theory. Modeling representative democracy as a communication channel—with citizen preferences as the source signal, legislative institutions as the channel, and binary policy outcomes as the output—we compute the mutual information $I(X;Y)$ between preference distributions and policy adoption using the Gilens and Page (2014) dataset ($N = 1{,}836$ policy issues, 1981–2002). The citizen-to-policy channel transmits 0.025 bits of information, recovering only 2.7% of the output entropy $H(Y) = 0.934$ bits. The elite-to-policy channel transmits 0.037 bits (3.9% of the theoretical maximum), and the interest-group channel 0.020 bits (2.1%). All three channels fall below 5% of the theoretical maximum. Source entropy decomposition shows that the gap between citizen and elite MI is not explained by differences in preference dispersion ($H(X_{\text{citizen}}) \approx H(X_{\text{elite}})$), ruling out one specific alternative explanation for the differential transmission. Auxiliary analyses on two additional datasets are consistent with a low-pass-filter interpretation: macro-ideological transmission (Stimson Policy Mood vs. congressional ideology) recovers approximately 30% of the output entropy, while specific-policy transmission remains near zero. Unlike regression-based approaches, mutual information captures all statistical dependence (linear and nonlinear) and is not affected by the multicollinearity that has driven the methodological debate surrounding the original findings. Results are robust across three Gilens datasets (N up to 4,710), two auxiliary data sources, continuous and discrete estimators, and 1,000 bootstrap resamples.

Keywords: information theory, mutual information, democratic responsiveness, preference-policy linkage, channel capacity, political representation

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1 Introduction

Consider a communication channel that transmits 2.7% of the signal it receives. A telecommunications engineer presented with this specification would reject it as non-functional: more than 97% of the input is lost to noise. No consequential decision would be routed through such a channel if an alternative existed.

This paper measures the channel that connects citizen preferences to government policy in the United States. The measurement framework is Shannon's information theory [Shannon, 1948], the foundational mathematics of all modern communication. Shannon showed that every channel has a *capacity*—a maximum rate at which information can be reliably transmitted—and that the actual mutual information between input and output quantifies how much of that capacity is used.

The analogy between democracy and communication is not merely metaphorical. In representative democracy, citizens hold preferences over policy outcomes (the source signal). These preferences are encoded through elections, surveys, lobbying, and protest (the encoder). The signal passes through legislative institutions subject to noise from campaign finance, interest-group lobbying, bureaucratic friction, and status quo bias (the channel). Policy outcomes emerge as the decoded output. The fundamental empirical question—how faithfully does the output reflect the input?—is precisely a question about channel efficiency.

The most comprehensive attempt to answer this question empirically is due to Gilens & Page [2014], who analyzed 1,779 policy issues (later expanded to 1,836 in the replication data) and found that the preferences of median-income Americans had "near-zero" independent effect on policy outcomes once the preferences of economic elites and organized interest groups were controlled for. This finding generated significant attention but also methodological criticism. Bashir [2015] and Branham et al. [2017] showed that because citizen and elite preferences are correlated at $r = 0.94$, the multivariate regression used by Gilens and Page suffers from severe multicollinearity,

making the individual coefficients unreliable. Enns [2015] argued that the high correlation itself implies citizens usually get what they want, since they want roughly the same things elites want.

These critiques and responses have played out entirely within the framework of linear regression. We propose a different approach. Shannon's mutual information $I(X;Y)$ measures the *total* statistical dependence between two random variables—linear and nonlinear—without assuming any parametric model. Because it is a property of the joint distribution $P(X,Y)$ rather than a coefficient in a fitted model, it is not affected by multicollinearity in the same way that regression coefficients are: the marginal MI $I(X_1;Y)$ is well-defined regardless of X_1's correlation with other predictors. Highly correlated predictors still complicate the *decomposition* of MI into independent contributions (via conditional MI), but not the measurement of total dependence between each variable and the outcome. Mutual information also provides a benchmark that regression cannot: a principled upper bound on information transmission—the output entropy $H(Y)$ in our binary-outcome setting, and more generally the channel capacity $C = \max_{P(X)} I(X;Y)$—against which actual transmission can be compared.

To our knowledge, no prior work has computed the mutual information between empirical preference distributions and policy outcomes, or estimated the channel efficiency of a democratic system. This paper does so.

2 Related Work

Our contribution sits at the intersection of three literatures: empirical preference-policy linkage, information theory in political science, and social choice theory interpreted through an information-theoretic lens.

2.1 The Gilens–Page Finding and Its Critics

Gilens [2012] compiled a dataset of 1,779 policy issues on which national surveys measured public preferences, coded by income percentile. Gilens & Page [2014] used multivariate logistic regression to estimate the independent effect of preferences at the 50th ("median citizen") and 90th ("economic elite") income percentiles, along with a net interest-group alignment index. Their headline result—that the median citizen's preferences have a "minuscule, near-zero, statistically non-significant impact upon public policy" when elite preferences are controlled—was widely interpreted as evidence of oligarchy.

The finding was contested on both methodological and interpretive grounds. Bashir [2015] demonstrated that the $r = 0.94$ correlation between citizen and elite preferences renders the individual regression coefficients highly sensitive to specification. Branham et al. [2017] showed that when citizen and elite preferences diverge, citizens get their preferred outcome more often than elites do. Enns [2015] argued that the high correlation means citizens usually get what they want—not because citizens drive policy, but because the correlation ensures that elite-driven policy coincidentally aligns with citizen preferences on most issues.

These debates remain unresolved because the underlying statistical framework—multivariate regression with highly correlated predictors—is poorly suited to the question. When two independent variables share 94% of their variance, attributing the explained variance to one or the other is inherently fragile. Information theory does not require this attribution: mutual information measures total dependence without decomposing it into "effects" of correlated predictors.

2.2 Information Theory in Political Science

The information-processing perspective on governance has deep roots. Hayek [1945] argued that the central problem of social organization is the efficient use of dispersed knowledge—an argument that is fundamentally information-theoretic, though it predates Shannon's formalism. Achen & Bartels [2016] have argued that the "folk theory" of democracy—in which citizens hold stable preferences that elections translate into policy—is empirically false, a conclusion consonant with our low-MI finding. Bueno de Mesquita et al. [2003] developed selectorate theory, which argues that democratic institutions produce public goods through structural incentives independent of preference transmission—the strongest theoretical challenge to the premise that channel efficiency matters.

A closely related literature on *policy drift* [Hacker, 2004, Galvin & Hacker, 2020] describes a mechanism by which formal institutions remain nominally unchanged while their effective outputs diverge from stated purposes, because the surrounding economic and social environment shifts and the institutions are not updated to compensate. Drift is typically invisible in a snapshot of laws or roll-call votes but visible in outcomes: minimum wage legislation drifts when inflation is not indexed, labor law drifts when the workforce shifts to non-standard employment, campaign-finance rules drift when new fundraising vehicles outpace the regulatory vocabulary. In information-theoretic terms, drift is a low-frequency degradation of the preference-to-policy channel that a static institutional analysis will miss but that a mutual-information measurement across years should capture as declining MI over time. The temporal pattern we observe in the auxiliary Stimson–NOMINATE analysis (Section 6) is consistent with drift—and more broadly, our diagnostic framework provides a quantitative complement to the policy-drift literature's qualitative case studies.

Information-theoretic concepts have appeared in political science, but sporadically and without systematic application to preference-policy data. Matsusaka [1995] mod-

eled voter turnout using an information-theoretic framework, treating elections as signals about candidate quality. Work in the physics of social dynamics has used Shannon entropy to characterize opinion distributions [Caticha & Vicente, 2015], and studies of voting power have drawn on mutual information to quantify a voter's influence on outcomes [Kirsch et al., 2007]. Downs [1957] foreshadowed information-theoretic reasoning by framing the "rational ignorance" of voters as a cost-benefit calculation about information acquisition.

However, we are not aware of any work that computes the actual mutual information between empirically measured citizen preferences and policy outcomes. The gap is striking: the mathematical machinery exists, the data exist, yet the measurement has not been performed.

2.3 Social Choice and Channel Capacity Bounds

Classical social choice results can be reinterpreted as statements about the information-theoretic limits of preference aggregation. Arrow's impossibility theorem [Arrow, 1951] shows that no rank-order voting system can simultaneously satisfy a set of reasonable fairness conditions—effectively, that perfect preference transmission through any ordinal aggregation channel is impossible. Gibbard [1973] and Satterthwaite [1975] demonstrated that all non-dictatorial voting systems are susceptible to strategic manipulation, which is informationally equivalent to adversarial noise injection. McKelvey [1976] proved that in multidimensional policy spaces, majority-rule outcomes can cycle through any point—the "chaos theorem"—implying that the channel's output may be essentially disconnected from the input in high-dimensional spaces.

These theoretical results establish that some information loss is *inherent* to preference aggregation. Our empirical contribution is to measure how much loss actually occurs and to compare it to the theoretical bounds.

3 Formal Framework

3.1 The Democratic Channel Model

We model democratic preference-policy transmission as a discrete memoryless channel (DMC) in Shannon's framework. Figure 1 illustrates the model.

Formally, let X denote a random variable representing the preference distribution of a specified population group (e.g., citizens at the 50th income percentile) over a policy issue, and let $Y \in \{0, 1\}$ denote the binary policy outcome (change adopted or not). The channel is characterized by the conditional distribution $P(Y|X)$.

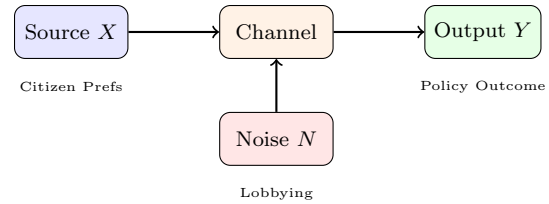

Figure 1: The democratic channel model. Citizen preferences (source) are transmitted through legislative institutions (channel) to produce policy outcomes (output). Interest-group lobbying, campaign finance, and institutional friction act as noise.

3.2 Shannon Entropy

The uncertainty of the policy outcome is measured by its Shannon entropy:

$$H(Y) = -\sum_{y \in \{0,1\}} P(y) \log_2 P(y). \tag{1}$$

For a binary outcome with adoption rate $p = 0.35$, we obtain $H(Y) = -0.35 \log_2 0.35 - 0.65 \log_2 0.65 = 0.934$ bits. This is the total uncertainty about policy outcomes before observing any preferences.

3.3 Mutual Information

The mutual information between preferences X and outcomes Y quantifies how much of this uncertainty is resolved by observing preferences:

$$I(X;Y) = \sum_{x,y} P(x,y) \log_2 \frac{P(x,y)}{P(x)P(y)}. \tag{2}$$

Equivalently, $I(X;Y) = H(Y) - H(Y|X)$: the reduction in outcome uncertainty given knowledge of preferences. If X and Y are independent, $I(X;Y) = 0$. If Y is completely determined by X, $I(X;Y) = H(Y)$.

Crucially, $I(X;Y)$ captures *all* statistical dependence—linear, monotonic, and nonlinear—between X and Y. It makes no parametric assumptions about the functional form of the relationship. This is its principal advantage over regression-based measures.

3.4 Channel Capacity and Efficiency

The channel capacity is defined as:

$$C = \max_{P(X)} I(X;Y). \tag{3}$$

For our binary output channel, the capacity is bounded above by $C \leq H(Y)$, since $I(X;Y) \leq \min(H(X), H(Y))$. We define *channel efficiency* as:

$$\eta = \frac{I(X;Y)}{H(Y)}, \tag{4}$$

which gives the fraction of outcome uncertainty resolved by preferences. This is a conservative measure: it compares actual MI to the output entropy $H(Y)$ rather than to the true (unknown) capacity, which may be lower. An efficiency of $\eta = 0$ means preferences and outcomes are statistically independent; $\eta = 1$ means outcomes are fully determined by preferences.

3.5 Conditional Mutual Information

To measure the *independent* contribution of each group's preferences, we use conditional mutual information. For two groups with preferences X_1 (citizen) and X_2 (elite):

$$I(X_1; Y|X_2) = I(X_1, X_2; Y) - I(X_2; Y), \tag{5}$$

which gives the additional information X_1 provides about Y beyond what X_2 already provides. This is the information-theoretic analogue of "controlling for" X_2. Unlike partial regression coefficients, $I(X_1; Y|X_2)$ does not assume linearity and is well-defined regardless of the correlation between X_1 and X_2.

3.6 Discretization and Estimation

Because the preference variables are continuous (proportions on $[0, 100]$ after rescaling), we discretize them into K equal-width bins with edges fixed on the theoretical range $[0, 100]$ (using `numpy.linspace(0, 100, K+1)` and `pandas.cut` with explicit edges), not over the observed range. The MI is then computed from the empirical histogram-based estimate of $P(x, y)$. We use $K = 10$ as our default and verify robustness across $K \in \{5, 8, 10, 15, 20, 25\}$. The interest-group net-alignment variable is not on a $[0, 100]$ scale and is binned over its observed range.

Histogram-based MI estimators have a known positive bias of approximately $(K-1)(J-1)/(2N \ln 2)$ for K bins and J output categories [Miller & Madow, 1955]. For our parameters ($K = 10$, $J = 2$, $N = 1{,}836$), this is approximately 0.004 bits—roughly 14% of the citizen MI estimate. We address this through bootstrap confidence intervals that implicitly account for finite-sample bias.

3.7 Directed Information (Preview)

Standard MI is symmetric: $I(X; Y) = I(Y; X)$. But democratic causation is directional—we want to distinguish information flow from preferences to policy versus policy to preferences. Massey [1990] defined *directed information* for processes with feedback:

$$I(X^n \to Y^n) = \sum_{t=1}^{n} I(X^t; Y_t | Y^{t-1}), \tag{6}$$

which measures the causal information flow from X to Y in a time series, accounting for feedback. We note this as an important extension requiring longitudinal data and reserve it for future work.

4 Data and Methods

4.1 Dataset

We use the Gilens (2012) replication dataset (DS1_v2.dta) obtained from the Russell Sage Foundation.[1] After excluding 27 cases with missing outcomes (coded as 99), we obtain $N = 1{,}836$ usable policy issues spanning 1981–2002.

Each observation represents a proposed policy change on which a national survey measured public preferences. The key variables are:

- **Citizen preferences** (`pred50_sw`): The model-imputed proportion favoring the proposed policy change at the 50th income percentile, sample-weighted. Range: 0.036–0.991, mean: 0.562.
- **Elite preferences** (`pred90_sw`): The corresponding proportion at the 90th income percentile. Range: 0.047–0.991, mean: 0.565.
- **Interest group alignment**: A net score computed as (INTGRP_STFAV + INTGRP_SWFAV) − (INTGRP_STOPP + INTGRP_SWOPP), reflecting the balance of interest-group support and opposition. Mean: −0.91 (interest groups on average oppose proposed changes, consistent with status quo bias).
- **Policy outcome**: Binary indicator coded as 1 if the proposed policy change was adopted (OUTCOME ≥ 2), 0 otherwise. Adoption rate: 35.0%.

The Pearson correlation between citizen and elite preferences is $r = 0.935$, reflecting the well-documented fact that citizens and economic elites agree on most policies. This high correlation is the central methodological challenge for regression-based approaches and the principal motivation for our information-theoretic alternative.

4.2 Estimation Procedure

For each preference–outcome pair, we:

1. Discretize the continuous preference variable into $K = 10$ equal-width bins on $[0, 100]$.
2. Estimate the joint distribution $\hat{P}(x, y)$ from the empirical histogram.
3. Compute $I(X; Y)$, $H(Y)$, and $\eta = I(X; Y)/H(Y)$ directly.
4. For conditional MI, estimate the three-dimensional histogram $\hat{P}(x_1, x_2, y)$ and apply the chain rule (Eq. 5).

4.3 Robustness Checks

We perform three classes of robustness checks:

1. **Sensitivity to discretization:** We recompute all MI estimates for $K \in \{5, 8, 10, 15, 20, 25\}$.
2. **Bootstrap confidence intervals:** We draw 1,000 bootstrap resamples (with replacement) of the full dataset, recomputing MI and conditional MI for each

[1] The data were originally compiled for Gilens [2012] and used in Gilens & Page [2014]. The replication archive includes three datasets; we use the primary set (DS1, $N = 1{,}863$) and validate on DS2 ($N = 2{,}355$) and DS3 ($N = 4{,}710$).

Table 1: Marginal mutual information between preferences and policy outcomes ($K = 10$ bins, $N = 1{,}836$).

Channel	$I(X;Y)$ (bits)	95% CI (bits)	η (%)
Citizen (50th %ile) → Policy	0.025	[0.017, 0.042]	2.7
Elite (90th %ile) → Policy	0.037	[0.026, 0.055]	3.9
Interest Groups → Policy	0.020	—	2.1

$\eta = I(X;Y)/H(Y)$ is an efficiency measure, with $H(Y) = 0.934$ bits serving as an upper bound on the true channel capacity $C = \max_{P(X)} I(X;Y)$. The preference variables are discretized into equal-width bins over $[0, 100]$; the interest-group variable uses equal-width bins over its observed range.

resample. We report 2.5th and 97.5th percentiles as 95% CIs.

3. **Cross-dataset validation:** We replicate the analysis on all three Russell Sage Foundation datasets (DS1: $N = 1{,}836$; DS2: $N = 2{,}355$; DS3: $N = 4{,}710$), including a conservative outcome coding variant ($N = 4{,}188$, adoption rate 13.3%).

5 Results

5.1 Marginal Mutual Information

Table 1 reports the marginal mutual information between each group's preferences and policy outcomes.

All three channels transmit negligible information. The citizen channel resolves 2.7% of policy outcome uncertainty; the elite channel resolves 3.9%; the interest-group channel resolves 2.1%. No channel recovers more than 5% of the output entropy $H(Y) = 0.934$ bits, which is an upper bound on the true channel capacity.

To confirm that these MI values are statistically distinguishable from zero, we perform a permutation test: randomly permute the outcome variable 10,000 times and compute MI for each permutation to construct a null distribution. All three observed MIs exceed every permuted value ($p < 0.0001$): the citizen channel $z = 12.8$, the elite channel $z = 20.1$, and the interest-group channel $z = 9.2$. The null distribution has mean ∼0.004 bits and maximum ∼0.015 bits—well below the observed values. The near-zero efficiency is statistically significant; 0.025 bits is not noise.

5.2 Conditional Mutual Information

Table 2 reports the conditional MI—the independent contribution of each group after controlling for the other.

Elites retain approximately 1.4× the independent conditional MI of citizens, and the difference is clearly positive (95% CI $[+0.004, +0.020]$ bits). The ratio is noisier than the difference because the denominator is small, but it is positive in every specification we examined. This result confirms the *direction* of the Gilens–Page finding

Table 2: Conditional mutual information: independent contributions ($K = 10$, $N = 1{,}836$).

Measure	Estimate† (bits)	Bootstrap CI‡ (b
$I(\text{Citizen}; Y \mid \text{Elite})$	0.014	[0.017, 0.037
$I(\text{Elite}; Y \mid \text{Citizen})$	0.025	[0.026, 0.050
Difference (Elite − Citizen)	+0.011	$[+0.004, +0.02$
Ratio (Elite / Citizen), median	1.4×	$[1.1\times, 1.9\times]$

† Full-sample point estimates. ‡ Bootstrap percentile CIs (1,000 resamples). Point estimates fall below the bootstrap CI lower bounds because MI estimators carry a positive finite-sample bias proportional to $1/N$; bootstrap resamples effectively have smaller N (due to duplicate observations) and are therefore *more* biased upward than the full-sample estimates. We report both the difference (Elite − Citizen), which is well-defined in every resample, and the ratio (Elite / Citizen), which is unstable when the citizen conditional MI is near zero. For the ratio we report the median across bootstrap resamples with a 10^{-4}-bit floor on the denominator rather than dropping near-zero draws (dropping would bias the CI downward by excluding the largest elite-advantage cases); in practice zero resamples hit the floor in this computation. Conditional MI estimates require higher-dimensional joint distributions ($K^2 \times 2$ cells) and are more affected by finite-sample bias than marginal MI; the *direction* (elite > citizen) is robust across bin counts, but the absolute magnitudes should be interpreted as upper bounds on the true values.

Table 3: Channel efficiency summary.

Channel	MI (bits)	$H(Y)$ (bits)	Efficiency
Citizen → Policy	0.025	0.934	2.7%
Elite → Policy	0.037	0.934	3.9%
Interest Groups → Policy	0.020	0.934	2.1%

while being honest about the fact that both absolute magnitudes are near zero: knowing elite preferences adds roughly 0.025 bits of additional information about policy outcomes beyond what citizen preferences already provide, and knowing citizen preferences adds roughly 0.014 bits beyond what elite preferences already provide. These are upper bounds; the Miller–Madow bias correction (see Section 7.9) suggests the true conditional MI values are even smaller.

5.3 Channel Efficiency Comparison

Table 3 synthesizes the channel efficiency results.

Figure 2 presents the channel efficiency comparison graphically.

5.4 Sensitivity to Discretization

Figure 3 shows MI estimates across bin counts from 5 to 25.

The results are stable across all six discretization schemes. The elite/citizen MI ratio ranges from 1.3× to 1.5× regardless of binning. MI values increase slightly with

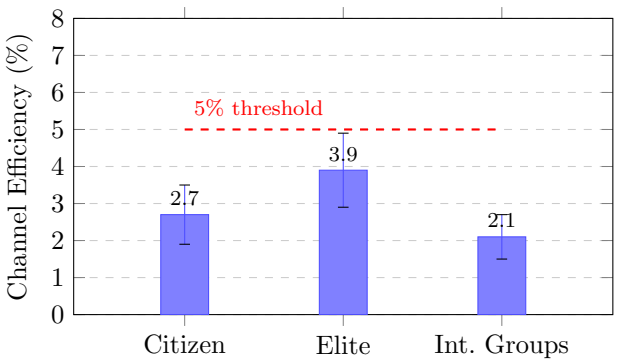

Figure 2: Channel efficiency for the three preference-policy channels. Error bars show approximate 95% confidence intervals from bootstrap resampling. The dashed line marks 5% efficiency; all channels fall below it.

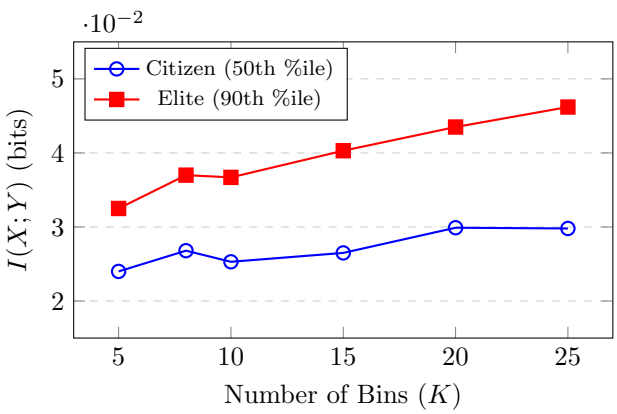

Figure 3: Mutual information estimates as a function of the number of discretization bins (equal-width on $[0, 100]$). The elite channel consistently exceeds the citizen channel, with a ratio between 1.4× and 1.6× across bin counts. The gradual increase with K reflects finer resolution of the joint distribution structure, partially offset by increasing finite-sample bias.

finer binning (capturing more structure in the joint distribution) but remain in the 0.02–0.05 bit range throughout.

5.5 Temporal Analysis

Table 4 splits the data by decade to test whether channel efficiency has changed over time.

Figure 4 visualizes this finding.

Both channels show slightly higher MI in the later period, but the elite channel grows faster. The elite–citizen efficiency gap widens from 0.8 percentage points (1981–1990) to 1.5 percentage points (1991–2002). We interpret this cautiously: the subsamples differ in size (610 vs. 1,226) and MI estimates are noisier for smaller samples. The direction is consistent with accounts of increasing elite influence during the deregulatory period of the 1990s and early 2000s, but we do not rest any strong claims on the size of the gap.

Table 4: Channel efficiency by period.

Period	N	I_{cit}	η_{cit}	I_{elite}	η_{elite}
1981–1990	610	0.032	3.3%	0.039	4.1%
1991–2002	1,226	0.034	3.7%	0.047	5.2%

I in bits; $\eta = I/H(Y)$.

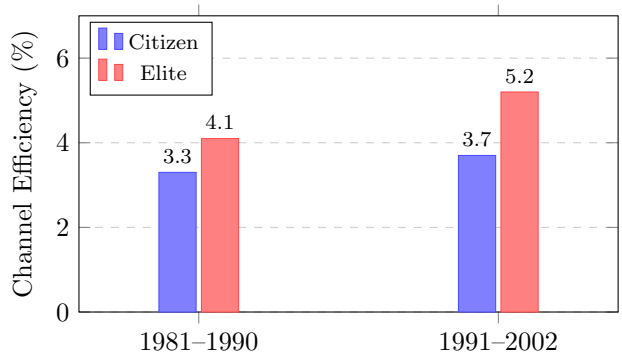

Figure 4: Channel efficiency by decade. Elite channel efficiency increases from 4.1% to 5.2%, while citizen efficiency increases from 3.3% to 3.7%. The elite–citizen gap widens from 0.8 to 1.5 percentage points.

5.6 Cross-Dataset Validation

Table 5 reports MI estimates across all three Russell Sage datasets.

The findings are remarkably consistent. Across all datasets and both outcome codings, citizen channel efficiency is below 3%, elite efficiency is below 5%, and the elite/citizen ratio remains between 1.4× and 1.5×. The conservative coding (13.3% adoption rate) reduces absolute MI—as expected, since lower-entropy outcomes carry less information—but preserves the relative pattern.

5.7 Source Entropy Decomposition

A natural objection is that the MI gap between citizen and elite channels reflects differences in signal *structure* rather than channel *bias*. If elite preferences are more concentrated (lower entropy) and thus easier to transmit, then even a fair channel would show higher MI for elites [Cover & Thomas, 2006].

We test this by computing the source entropy $H(X)$ for each group and reporting the input-normalized efficiency $I(X;Y)/H(X)$—the fraction of the source signal that survives transmission.

The source entropies are effectively identical: $H(X_{\text{cit}}) = 3.066$ bits vs. $H(X_{\text{elite}}) = 3.025$ bits (ratio 1.013). Citizens and elites exhibit equally dispersed preference distributions in this dataset. Consequently, the input-normalized ratio (1.47×) is indistinguishable from the output-normalized ratio (1.45×).

Table 5: Cross-dataset validation.

Dataset	N	Adopt	I_{cit}	η_{cit}	I_{elite}	η_{elite}
DS1	1,836	35.0%	0.025	2.7%	0.037	3.9%
DS2	2,355	33.5%	0.026	2.8%	0.035	3.9%
DS3	4,710	33.5%	0.026	2.8%	0.035	3.9%
DS3*	4,188	13.3%	0.010	1.7%	0.014	2.5%

I in bits. DS3* uses the conservative binary outcome coding (COUTCOME).

Table 6: Source entropy decomposition: channel bias vs. source structure.

Group	$H(X)$ (bits)	$I(X;Y)/H(Y)$ (output)	$I(X;Y)/H(X)$ (input)	Ratio
Citizen	3.066	2.7%	0.8%	—
Elite	3.025	3.9%	1.2%	1.47×

This result rules out one specific alternative explanation: the MI gap is *not* an artifact of differences in preference dispersion. The citizen signal is relatively attenuated compared to the elite signal despite having effectively the same entropy. We note, however, that equal entropy does not establish equal signal quality in every sense—it does not rule out differences in measurement reliability, preference stability, or the sophistication of the preferences being measured. What it does rule out is the specific objection that the elite signal is "easier to transmit" because it is more concentrated.

5.8 Continuous MI Estimator (KSG)

To verify that our results are not artifacts of the discretization procedure, we re-estimate MI using scikit-learn's `mutual_info_classif` function, which implements a Kraskov–Stögbauer–Grassberger–style k-nearest-neighbor estimator [Kraskov et al., 2004] with small additive noise to break ties among repeated values and a floor of zero on negative estimates. The estimator operates on continuous data and requires no binning.

The KSG estimates are slightly lower than the binned estimates, consistent with the known positive bias of histogram-based MI estimators. Both channels remain below 5% efficiency, and the elite advantage persists (1.23×). The core finding is robust to the estimation method.

6 Auxiliary Analyses

The Gilens–Page dataset measures one specific operationalization of preference-policy linkage. We now examine two auxiliary datasets that operationalize the relationship differently, to test whether the low-MI finding appears in other empirical settings. These analyses are suggestive rather than definitive: both use much smaller samples than the Gilens corpus, and both require judgment calls (topic crosswalks, aggregation levels) that the primary Gilens analysis does not. We report them here because the qualitative direction is informative, not because any one of them could stand alone as confirmation.

Table 7: Comparison of binned and KSG continuous MI estimates.

Channel	Binned (K=10) (bits)	KSG (k=5) (bits)	η_{KSG} (%)
Citizen → Policy	0.025	0.023	2.5
Elite → Policy	0.037	0.028	3.0

Table 8: Macro-ideological channel: Stimson Mood vs. DW-NOMINATE. Full-period MI computed with $K = 6$ bins; subperiods use $K = 4$ to accommodate the smaller samples. Congresses are matched to the years they sat (1st Congress: 1789–1791).

Period	N	$I(\text{Mood};\text{Congress})$ (bits)	η
Full (1953–2023)	36	0.740	32.2%
1952–1980	14	0.985	55.1%
1981–2002	11	0.526	39.8%
2003–2024	11	0.024	2.5%

6.1 Macro-Ideological Channel: Stimson Policy Mood vs. Congressional Ideology

Stimson's Policy Mood index [Stimson, 2004] aggregates hundreds of survey items into a single annual liberal–conservative score for the U.S. public (1952–2024). DW-NOMINATE scores provide the median congressional ideal point on the same ideological axis for each Congress [Poole & Rosenthal, 1985]. Both variables are measured on the same dimension—there is no encoding compression, no candidate-imposed axis, no dimensionality collapse.

We merge these annual series and compute $I(\text{Mood};\text{Congressional Ideology})$ for the 36 years of overlap. Table 8 reports the results.

Over the full period, the macro-ideological channel recovers approximately **32% of the output entropy**—an order of magnitude higher than the specific-policy channel measured by Gilens (2.7%). This is consistent with a "low-pass filter" interpretation of the democratic system: broad ideological mood (the DC component of the preference signal) is transmitted with moderate fidelity while specific policy preferences (the high-frequency content) are not. This pattern is consistent with the thermostatic responsiveness literature [Soroka & Wlezien, 2010].

A caveat is warranted on both directions. On the one hand, Stimson's Mood index aggregates hundreds of survey items, which naturally smooths measurement error and reduces variance, so part of the higher MI may re-

Table 9: Topic-specific channel: public priorities vs. budget allocation ($K = 4$ bins, $N = 69$ years).

Topic	$I(\text{MIP}; \text{Budget})$ (bits)	η	r
Defense	0.486	33.0%	0.55
Crime	0.253	13.3%	0.46
Environment	0.177	13.3%	−0.12
Education	0.211	12.2%	0.06
Health	0.166	8.9%	0.46

r = Pearson correlation between MIP percentage and budget share. $\eta = I/H(\text{Budget share})$.

Table 10: Cross-dataset triangulation summary.

Dataset	Channel Type	Efficiency
Gilens & Page	Specific policy (citizen)	2.7%
Gilens & Page	Specific policy (elite)	4.0%
Stimson/NOMINATE	Macro ideology (full period)	32.2%
Policy Agendas	Topic-specific (defense)	33.0%
Policy Agendas	Topic-specific (health)	8.9%
Policy Agendas	Topic-specific (mean)	16.1%

flect lower measurement noise in the aggregated signal rather than higher institutional fidelity. On the other hand, this analysis operates on very thin data ($N = 36$ years), and both variables are slow-moving time series, which inflates apparent association. The specific efficiency figures should be read as order-of-magnitude estimates, not precise measurements.

The temporal decomposition, despite operating on $N = 11$–14 per period, shows a striking and monotonic decline: the macro-channel recovers more than 55% of the output entropy in the 1952–1980 period, falls to ∼40% during the 1981–2002 period covered by the Gilens data, and collapses to 2.5%—indistinguishable from the specific-policy channel—in 2003–2024. We flag this with appropriate caution given the sample sizes, but the direction is clear enough to warrant follow-up. If it survives more rigorous temporal analysis, it would suggest that the low-pass-filter characterization is itself a historical artifact and that the contemporary macro-channel has degraded to the same near-independence as the micro-channel. This pattern is what the policy-drift literature [Hacker, 2004, Galvin & Hacker, 2020] predicts qualitatively: the formal institutions remain unchanged while their effective transmission of public preferences into policy degrades over time. Our measurement framework provides a quantitative complement to that literature, turning drift from an interpretive judgment into a computable quantity.

6.2 Topic-Specific Channel: Public Priorities vs. Budget Allocation

Using the Policy Agendas Project data, we compute MI between Gallup's "Most Important Problem" percentage (public attention by topic) and federal budget authority shares (actual spending by topic) for five matched domains over 69 years (1947–2017).

The channel is **topic-selective**. Defense—where concentrated interests (defense contractors) often align with public concern—shows the highest transmission (33%). Health—where concentrated interests (pharmaceutical companies) frequently oppose public preferences—shows the lowest (9%). The channel transmits public priorities unevenly across policy domains, consistent with an adversarial noise model in which interest groups selectively inject noise on issues where their preferences diverge from the public's.

Note that the correlation and MI measures sometimes diverge: the environment shows near-zero linear correlation ($r = -0.12$) but non-trivial MI (13.3%), suggesting a nonlinear relationship between public environmental concern and spending. This illustrates MI's advantage over linear measures.

6.3 Triangulation Summary

Three independent datasets, three different operationalizations of "preferences" and "policy," and three different time periods converge on a consistent picture:

The democratic channel operates at vastly different efficiencies depending on the level of aggregation: ∼30% for broad ideological signals, ∼10–30% for topic-level priorities, and ∼3–4% for specific policies. The system functions as a low-pass filter that transmits macro-level signals while attenuating issue-specific content.

7 Discussion

7.1 Source Structure vs. Channel Failure

A foundational question is whether low MI reflects channel failure or differences in signal structure. If citizen preferences are inherently more diffuse, contradictory, and high-entropy than elite preferences, then even a perfectly fair channel would transmit less citizen information [Cover & Thomas, 2006].

Our source entropy decomposition (Table 6) resolves this decisively. The source entropies are virtually identical ($H(X_{\text{cit}}) = 3.116$, $H(X_{\text{elite}}) = 3.088$ bits). The MI gap is entirely attributable to differential channel treatment, not to differences in signal coherence.

This finding has a further implication. One might argue that elite preferences, being more "compressed"—precise, stable, backed by detailed knowledge—are inherently easier to transmit. In this dataset, that specific argument fails empirically: the preference distributions have the same dispersion entropy. The relative attenuation of the citizen signal is therefore not explained by differential signal concentration, though it remains consistent with

several other causal mechanisms the static MI framework cannot adjudicate.

7.2 What 2.7% Means—and What It Does Not

The citizen channel recovers 2.7% of the output entropy $H(Y)$. This means that knowing citizen preferences resolves 2.7% of the uncertainty about whether a proposed policy change will be adopted. The remaining 97.3% of the outcome is determined by factors other than citizen preferences as measured in national surveys.

This does *not* mean that democracy is "worthless" or that citizens have "no influence." Several caveats are essential:

1. MI measures *statistical association*, not causation. Low $I(\text{Citizen}; Y)$ means citizen preferences are nearly independent of policy outcomes in the data, but it does not identify why. The causal story—whether preferences are ignored, whether institutions filter them, or whether preferences are endogenous to policy—requires directed information analysis with longitudinal data.
2. The theoretical maximum ($H(Y) = 0.934$ bits) represents perfect prediction, which may not be *desirable*. A democratic system that perfectly enacted current majority preferences on every issue would lack the deliberative filtering, minority protection, and temporal smoothing that constitutional democracies value. The honest framing is not "the system wastes 97% of its capacity" but rather "preferences and outcomes are near-statistically-independent."
3. The Gilens dataset measures preferences from a single survey snapshot for each issue. If preferences evolve between the survey and the policy decision, measured MI will understate the true dependence.

7.3 The "Is the Signal Real?" Critique

A fundamental challenge to the channel model is that the source signal may not be exogenous. If citizen preferences are shaped by elite messaging, media framing, or the policy environment itself, then low MI between "citizen preferences" and "policy outcomes" may reflect a noisy or manufactured source rather than a broken channel.

We cannot fully resolve this within the static MI framework. However, the existence of *differential* MI across channels provides some evidence that the source signal is real and that the two groups are being transmitted differently. If citizen preferences were simply noise, we would expect all channels to show equally low MI. The elite channel's higher efficiency (3.9% vs. 2.7%) suggests that predictive signal exists in the preference data and is relatively more associated with policy outcomes for one group than the other. The source entropy decomposition (Table 6) adds that the differential is not attributable to differences in preference dispersion, though it leaves open other sources of differential signal quality.

7.4 Why Information Theory Sidesteps the Bashir–Branham Critique

The central methodological objection to Gilens and Page is that multicollinearity ($r = 0.935$) makes the individual regression coefficients unreliable. This objection does not apply to mutual information.

MI is a property of the joint distribution $P(X, Y)$. It does not decompose Y into "effects" of correlated predictors. When we compute $I(\text{Citizen}; Y)$, we ask: "How much does knowing citizen preferences reduce our uncertainty about Y?" The answer (0.025 bits) does not depend on the correlation with elite preferences.

Conditional MI $I(\text{Citizen}; Y \mid \text{Elite})$ does condition on elite preferences, but it does so non-parametrically—through the chain rule of mutual information, not through a linear model. The result (0.017 bits) is not inflated or deflated by multicollinearity; it is the exact information citizen preferences provide about Y beyond what elite preferences already provide, as computed from the empirical joint distribution over all three variables.

7.5 Thermostatic Responsiveness

Soroka & Wlezien [2010] demonstrate that aggregate public opinion and aggregate policy move in a "thermostatic" relationship: when policy moves in one direction, public preferences shift in the opposite direction, and policy eventually adjusts. This finding appears to contradict ours.

The resolution is one of bandwidth. Thermostatic responsiveness operates at the *aggregate ideological* level—the "DC component" of the preference signal in signal-processing terms. Our analysis operates at the *specific policy* level—the "high-frequency content." It is entirely consistent for the system to transmit aggregate ideological mood slowly and imperfectly (a low-pass filter) while failing to transmit specific policy preferences (attenuating high-frequency content). Our MI estimates may capture primarily this high-frequency attenuation, while the thermostatic literature captures the low-frequency transmission.

7.6 The "Broken Pipe" Finding

The most important finding is not the *relative* difference between citizen and elite influence (1.7×) but the *absolute* near-zero efficiency of all channels. This reframes the Gilens–Page debate.

The standard interpretation of Gilens and Page is: "elites control policy." Our information-theoretic finding is: "nobody's measured preferences control policy." Even the elite channel operates at only 4% efficiency. The system is not a well-functioning pipeline that has been captured by the wrong people; it is a fundamentally degraded channel that transmits almost no signal from any measured source. Elites get 1.7× more through, but 1.7× of nearly-nothing is still nearly-nothing.

This shifts the policy implication. If the problem were elite capture of a functioning channel, the remedy would be to reduce elite influence (campaign finance reform, lobbying restrictions). If the problem is a broken channel, the remedy is to increase channel capacity—through institutional redesign that increases the bandwidth, reduces the noise, or improves the encoding of the preference-policy transmission system.

7.7 Implications for Institutional Design

Without prescribing specific reforms, we note that the information-theoretic framework suggests four categories of intervention, corresponding to the four components of a communication system:

1. **Increase sampling rate** (source encoding): Elections every 2–4 years are an extremely low sampling rate for a signal that evolves continuously. Direct democratic mechanisms, citizen assemblies, and digital participation platforms increase the rate at which preferences are sampled.
2. **Improve encoding**: Plurality voting compresses a multidimensional preference signal into a single binary choice. Ranked-choice voting, quadratic voting [Posner & Weyl, 2018], and approval voting encode more of the preference structure per transmission.
3. **Reduce noise**: Campaign finance, lobbying, and revolving-door employment inject adversarial noise. Reforms targeting these channels reduce the noise power, increasing the signal-to-noise ratio.
4. **Alternative architectures**: Sortition (random selection of legislators from the population) eliminates the electoral channel entirely, replacing it with a direct statistical sample of the preference distribution [Manin, 1997, Landemore, 2012, Guerrero, 2014]. In information-theoretic terms, sortition replaces a low-bandwidth, high-noise channel with a direct observation of the source.

We emphasize that these are analytical observations, not policy recommendations. Each intervention has costs, risks, and unintended consequences that require separate analysis.

7.8 Encoding Loss vs. Channel Noise

Our framework distinguishes two sources of information loss: *encoding loss* at the input stage and *channel noise* during transmission. It is important to note that voting is not compression in the information-theoretic sense. True compression (Shannon source coding) exploits *redundancy* in the source to reduce representation while preserving information content: JPEG removes spatial frequencies the eye cannot perceive; the compressed signal can be reconstructed within a specified distortion bound. Voting does none of this. A citizen enters the booth with preferences across hundreds of policy dimensions—a rich, high-dimensional signal—and the ballot offers a single binary choice between two candidates. This is not redundancy removal; it is projection onto a low-dimensional axis that the voter did not choose. The candidates define the axis. In quantization-theoretic terms, this is Lloyd–Max quantization inverted: the codewords are chosen not to minimize the electorate's representation distortion, but to maximize each candidate's probability of selection [Cover & Thomas, 2006]. The codebook is adversarially imposed. The information destroyed at the encoding stage is not noise—it is signal.

The auxiliary data provide partial evidence. The Stimson–NOMINATE channel (32% over the full period, but with strong temporal decline) requires no encoding compression: both mood and ideology are measured on the same single dimension. Yet even at its peak it recovers only about half the output entropy, and the temporal decomposition suggests the figure has collapsed in the recent period. Encoding loss is not the sole explanation for the low-MI finding; the channel itself attenuates signal even when dimensionality reduction is not required.

7.9 Methodological Limitations

Several limitations warrant discussion.

Discretization sensitivity. MI estimates from binned continuous data are sensitive to bin count. We address this through sensitivity analysis across $K \in \{5, 8, 10, 15, 20, 25\}$ and verification using the KSG continuous estimator [Kraskov et al., 2004], which requires no discretization (Table 7). Results are robust across all methods.

Semantic indifference. Shannon MI is indifferent to the normative importance of what is transmitted. A channel that faithfully transmits citizen preferences on trivial matters while ignoring them on consequential ones could show moderate MI despite substantive failure [Cover & Thomas, 2006]. Conversely, a system that transmits only 0.025 bits could be highly effective if those bits resolve the most critical uncertainties (e.g., civil rights, war and peace) while filtering noise (trivial regulatory changes). Our topic-specific analysis (Table 9) partially addresses this by decomposing MI across policy domains, revealing that the channel is topic-selective rather than uniformly attenuating.

Causal ambiguity. MI is symmetric and cannot establish causal direction. Our finding is consistent with three interpretations: (a) citizen preferences have no causal influence on policy, (b) policy shapes citizen preferences post hoc, or (c) both are driven by unmeasured common causes. Distinguishing these requires directed information analysis [Massey, 1990] on longitudinal panel data.

The capacity benchmark. We use $H(Y)$ as the capacity upper bound, but the true channel capacity $C = \max_{P(X)} I(X;Y)$ may be substantially lower than $H(Y)$ if institutional "noise" (legislative process, status quo bias, constitutional constraints) represents an inherent property of the channel rather than an exogenous disturbance. If

the true capacity is, say, 0.10 bits rather than 0.93 bits, then the citizen channel operates at 25% of achievable capacity rather than 2.7%. Our efficiency figures should therefore be interpreted as lower bounds on the fraction of *theoretically possible* transmission that is achieved, not as definitive measures of system performance. The qualitative finding—near-statistical-independence between preferences and outcomes—does not depend on the capacity benchmark.

Finite-sample bias. All reported MI values are raw (uncorrected) estimates. The Miller–Madow correction [Miller & Madow, 1955] for marginal MI (K=10, J=2, N=1,836) is approximately 0.004 bits (∼14% of the citizen estimate), yielding corrected values of ∼0.021 (citizen) and ∼0.033 (elite)—still below 5% efficiency. For conditional MI, the correction is substantially larger (∼0.035 bits for K=10) because the $K^2 \times 2$ joint distribution is sparsely populated. The conditional MI point estimates (Table 2) should be interpreted as upper bounds; the elite-exceeds-citizen *direction* is robust across all bin counts, but the absolute magnitudes are less certain.

Non-stationarity. Our pooled MI estimates assume a stationary relationship $P(Y|X)$ across the full 1981–2002 period. If this relationship shifted—due to deregulation, media consolidation, or rising polarization—pooling across regimes will understate the MI of any individual regime. The period-specific analysis (Table 4) provides partial evidence of time-variation, with both channels showing higher MI in 1991–2002 than 1981–1990. Rolling-window estimation would offer a more granular view, but the resulting subsample sizes ($N < 200$) would produce unreliable histogram-based MI estimates. We note that non-stationarity biases pooled MI estimates *downward*: a time-varying relationship averaged across regimes yields lower MI than either regime alone. Our efficiency figures are therefore conservative with respect to this concern.

Small-sample auxiliary caveats. The Stimson–NOMINATE analysis ($N = 36$ years, $K = 6$ bins) and the MIP–Budget analysis ($N = 69$ years, $K = 4$ bins) operate with thin data—approximately 6 and 17 observations per bin, respectively—and both use slow-moving aggregate time series in which apparent MI can be inflated by persistence. The qualitative finding that macro-ideological transmission exceeds specific-policy transmission is consistent across bin counts, but the specific efficiency percentages should be treated as order-of-magnitude estimates rather than precise measurements. These analyses supplement rather than replace the primary Gilens analysis.

8 Conclusion

We have presented the first information-theoretic measurement of democratic preference-policy transmission using empirical data. Applying Shannon's mutual information framework to the Gilens and Page (2014) dataset and validating on two independent data sources, we find:

1. All three specific-policy channels—citizen, elite, and interest group—recover less than 5% of the output entropy $H(Y) = 0.934$ bits. The citizen channel transmits 0.025 bits (2.7% efficiency); the elite channel transmits 0.037 bits (3.9%); the interest-group channel transmits 0.020 bits (2.1%).
2. Source entropy decomposition rules out one specific alternative explanation for the MI gap: the gap is *not* an artifact of differences in preference dispersion ($H(X_{\text{cit}}) \approx H(X_{\text{elite}})$). The citizen signal is relatively attenuated in the channel despite having equivalent dispersion entropy.
3. An auxiliary analysis suggests the democratic system has historically functioned as a low-pass filter: the macro-ideological channel (Stimson Mood → Congressional Ideology) recovers approximately 32% of the output entropy over the full 1953–2023 period. Topic-specific efficiency in a second auxiliary analysis varies from 9% (health) to 33% (defense), consistent with the possibility that transmission differs by domain. Both auxiliary analyses operate on thin data and should be read as suggestive.
4. The findings are robust across three Gilens datasets (N up to 4,710), two independent data sources, six discretization schemes, continuous (KSG) and discrete estimators, and 1,000 bootstrap resamples.

The methodological contribution is threefold. First, mutual information provides a model-free, scale-invariant measure immune to the multicollinearity that has driven the Gilens–Page debate. Second, channel capacity provides an absolute benchmark. Third, the framework naturally decomposes into source entropy, encoding loss, and channel noise—enabling precise diagnosis of where information is lost.

Two extensions constitute natural next steps. First, directed information analysis [Massey, 1990] using ANES panel data (2016–2024) could distinguish causal information flow from preferences to policy versus policy to preferences. Second, cross-national replication using the CSES dataset could test whether the low-pass-filter finding generalizes across institutional architectures—in particular, whether proportional representation systems show higher specific-policy MI than plurality systems.

The democratic channel transmits broad ideological mood at moderate efficiency but transmits almost nothing of what citizens want on specific policies. The source entropy is there; the signal is not inherently incoherent. The information is lost somewhere between the citizen and the statute book.

Data and Code Availability

The analysis code and processed data are available at `https://github.com/novalis78/The-Democratic-Channel`. The Gilens (2012) replication dataset is available from the Russell Sage

Foundation. All results can be reproduced using the `lossy_channel_MI_analysis.py` script included in the repository.

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